Latent Altruism: Steering Cooperative Intent in Large Language Models

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Abstract

Large Language Models (LLMs) are increasingly deployed as autonomous agents in multi-agent environments. However, without explicit alignment, these models often default to selfish or defection-oriented behaviors in game-theoretic settings. In this paper, we investigate the internal representation of strategic intent within LLMs and propose a zero-shot intervention method using Activation Engineering. By extracting a "Cooperation" Steering Vector from the Iterated Prisoner's Dilemma, we successfully shift a 32B-parameter LLM from a 2% baseline cooperation rate to 100%. We demonstrate that this vector generalizes zero-shot to other non-zero-sum games, achieving 97.8% cooperation in Stag Hunt and 100% in Chicken Game, preserving the model's core language capabilities (Perplexity Ratio ≈ 1.0). Through mechanistic interpretability techniques, we localize the formation of strategic intent to the mid-to-late layers (specifically Layer 57), where the linear separability of cooperative and selfish representations peaks. Finally, we expose a critical vulnerability: activation steering is overwhelmingly subjected to "Contextual Override" when faced with adversarial, competitive prompts. Our findings provide a geometric understanding of strategic behavior in LLMs and highlight the boundaries of activation interventions for AI safety.

1 Introduction

The deployment of Large Language Models (LLMs) in complex, multi-agent systems necessitates that these models not only perform reasoning but also engage in cooperative behaviors **?**. However, game-theoretic evaluations reveal that LLMs often default to Machiavellian or strictly self-maximizing strategies, failing to achieve Pareto-optimal outcomes in social dilemmas like the Prisoner's Dilemma **?**. Traditional methods to align these models, such as prompting, are heavily context-dependent and prone to adversarial exploitation **?**.

Recent advancements in Mechanistic Interpretability and Activation Engineering **??** suggest that complex high-level concepts, including moral and strategic intents, are encoded as linear directions within the latent representation space of neural networks. Building on this, we ask: *Can we identify and directly stimulate a generalized "cooperative intent" within an LLM without altering its weights or linguistic capabilities?*

In this work, we present a comprehensive study on the geometry, generalizability, and limitations of cooperative intent in a 32B-parameter Instruct model. Our contributions are four-fold:

1. **Geometric and Layer-wise Localization**: We show that cooperative and selfish intents are distinctly separated in the latent space (Silhouette score: 0.720). Furthermore, we localize the emergence of this strategic intent to Layer 57, suggesting a separation between strategic reasoning and vocabulary projection.

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2. **Cross-Game Generalization**: We demonstrate that a steering vector extracted solely from the Prisoner's Dilemma generalizes zero-shot to the Stag Hunt and Chicken games.
3. **Capability Preservation**: We compare Steering Vectors (SV) with Contrastive Activation Addition (CAA) and Representation Engineering (RepE), proving that SV achieves 100% cooperation with no degradation in perplexity.
4. **Vulnerability Analysis**: We expose the "Contextual Override" phenomenon, where strong adversarial prompts entirely nullify the effects of activation steering.

2 Related Work

Game Theory and LLMs Evaluating LLMs using classical game theory has become a standard methodology to probe their social intelligence. Previous works have simulated the Iterated Prisoner's Dilemma (IPD) to test LLM alignment, finding that models often exhibit greedy behaviors unless heavily prompted or fine-tuned **?**.

Activation Engineering Instead of modifying model weights via RLHF or fine-tuning, Activation Engineering adds bias vectors directly to hidden states during inference. **?** introduced Steering Vectors to alter model personas. **?** formalized Representation Engineering (RepE) using PCA on contrastive datasets. **?** proposed Contrastive Activation Addition (CAA) for behavioral alignment. Our work extends these techniques to multi-agent game theory, providing both structural localization and robustness benchmarking.

3 Methodology

3.1 Experimental Setup and Environments

We utilize the Qwen2.5-32B-Instruct model with 4-bit quantization. We evaluate the model's behavior across three classic non-zero-sum games: the Iterated Prisoner's Dilemma (IPD), the Stag Hunt, and the Chicken Game. We benchmark the model against 5 standard hard-coded opponents: Tit-For-Tat (TFT), Always Cooperate (AllC), Always Defect (AllD), Win-Stay-Lose-Shift (WSLS), and Random. Each game runs for 50 rounds and is repeated 5 times to compute 95% confidence intervals.

3.2 Extracting the Steering Vector

We collect the hidden states $H_l^{(i)}$ of the model at layer l for the final token of the prompt. We define two behavioral prompts: $\mathcal{P}_{AllC}$ (instructing the model to always cooperate) and $\mathcal{P}_{AllD}$ (instructing the model to always defect). The Steering Vector v_l for layer l is computed as the mean difference of the hidden states over N calibration rounds:

$$v_l = \frac{1}{N}\sum_{i=1}^{N} H_l(\mathcal{P}_{AllC}^{(i)}) - \frac{1}{N}\sum_{i=1}^{N} H_l(\mathcal{P}_{AllD}^{(i)}) \tag{1}$$

During inference, we intervene on the model's forward pass by adding the scaled vector to the target layer's hidden state: $\tilde{H}_l = H_l + \alpha \cdot v_l$, where α is the steering strength.

4 Experiments and Results

4.1 The Geometry of Strategic Intent

To understand how the model encodes strategies, we perform Principal Component Analysis (PCA) and t-SNE on the hidden states of the AllC, AllD, Baseline, and Steered conditions. As shown in Figure 1, at the last layer, the representations of AllC and AllD are linearly separable, achieving a Silhouette score of 0.720. Interestingly, the Baseline representations naturally cluster closer to AllD, confirming the model's default selfish bias. Upon applying the steering vector, the Steered trajectory is geometrically shifted from the AllD region strictly into the AllC region.

To determine the optimal intervention point, we scanned all 65 layers of the model, measuring the Fisher Discriminability Index between the AllC and AllD representations. As illustrated in Figure 2,

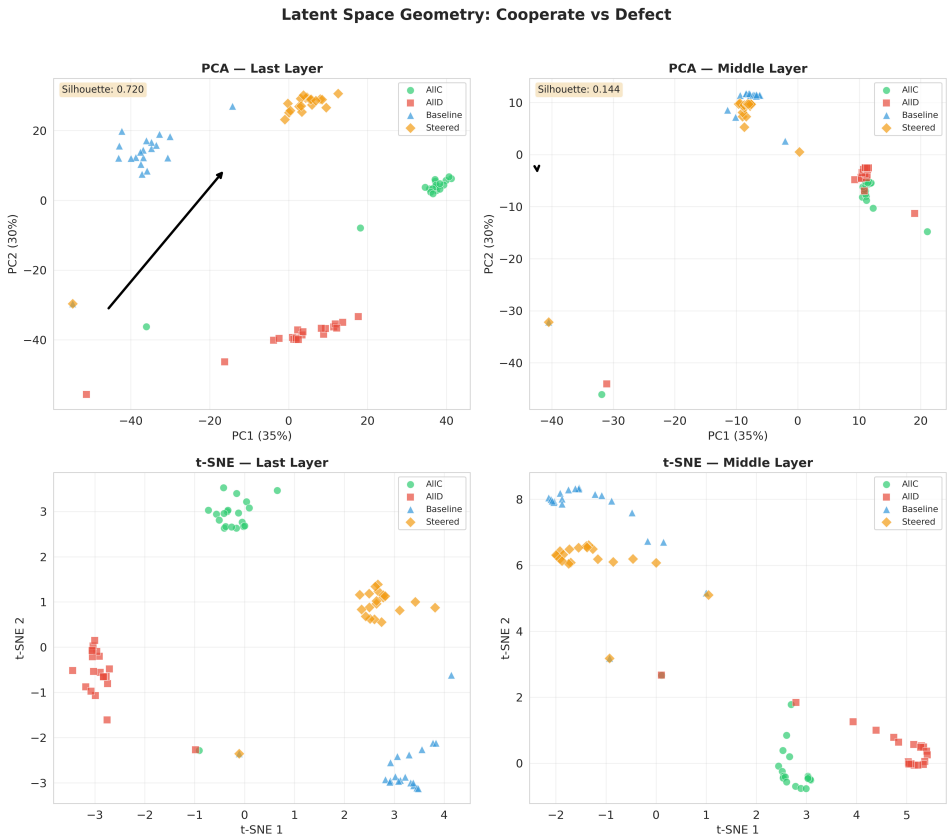

Figure 1: Latent Space Geometry of Cooperative vs. Defective intent. The Steered condition (yellow diamonds) is shifted by the vector, successfully overriding the Baseline's (blue triangles) default bias.

the discriminability does not peak at the final layer, but rather at **Layer 57**. This suggests a specific "strategic bottleneck" in the architecture: Layer 57 is where abstract strategic intent is maximally crystallized, before the subsequent layers project this intent into raw token probabilities.

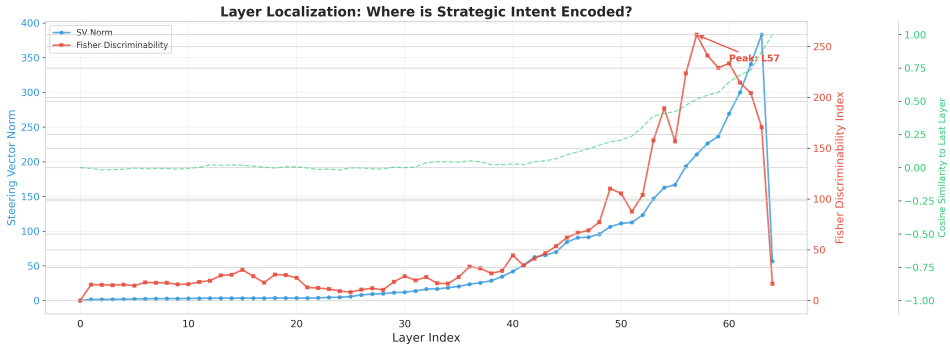

Figure 2: Layer Localization. The Fisher Discriminability (red) peaks at Layer 57, identifying the optimal layer for extracting and injecting the strategic steering vector.

4.2 Zero-shot Cross-Game Generalization

A core question is whether the model learns a superficial token correlation or a generalized concept of "cooperation." We extracted the steering vector entirely from the IPD environment and applied it to the Stag Hunt and Chicken Game. The results demonstrate remarkable zero-shot transferability. The steering vector achieved a 97.8% transfer rate in the Stag Hunt and 100% in the Chicken Game.

When facing the adversarial AllD opponent, standard prompting failed drastically (reverting to 46% and 4% cooperation, respectively), whereas the steered model sustained robust cooperation (94% and 82%), essentially exhibiting "blind altruism" despite receiving the lowest possible payoffs.

4.3 Method Comparison and Capability Preservation

We benchmarked simple Steering Vectors (SV) against Contrastive Activation Addition (CAA) and Representation Engineering (RepE). As shown in our evaluations, SV and CAA exhibit high sensitivity, shifting the model to 100% cooperation at a minimal intervention strength ($\alpha = 0.05$). RepE, conversely, exhibits high inertia, requiring $\alpha \geq 0.3$ to overcome the model's default bias.

Crucially, an alignment technique is only viable if it does not collapse the model's linguistic capabilities. We evaluated the Perplexity (PPL) ratio of the steered outputs relative to the unsteered baseline (Baseline PPL: 1.84). The SV method achieved a minimum PPL ratio of 1.0, indicating zero degradation in language modeling quality. CAA performed similarly (PPL ratio 1.004). This proves that localized activation engineering smoothly alters behavior without corrupting syntax or logic.

4.4 Vulnerability to Adversarial Contexts: Contextual Override

While activation steering is resilient against adversarial *opponents* (e.g., AllD), it exhibits critical brittleness against adversarial *contexts*. We tested the steered model using 4 adversarial system prompts designed to induce paranoia or dominance (e.g., "Intelligence reports show your opponent plans to betray you").

Despite the active presence of the steering vector, the recovery rate of cooperation was exactly 0%. The self-attention mechanism assigned overwhelmingly large weights to adversarial tokens (e.g., "betray", "maximize"), creating local activations that completely overrode the global geometric shift introduced by the steering vector. This "Contextual Override" phenomenon highlights a fundamental boundary condition: latent space interventions cannot easily bypass explicitly hostile contexts in the prompt manifold.

5 Discussion and Limitations

Our results confirm that abstract strategic concepts are linearly represented and manipulable in LLMs. However, the Contextual Override phenomenon presents a significant limitation for AI safety. If malicious users can inject adversarial personas into the prompt, activation engineering alone is insufficient to guarantee cooperative alignment. Future work must investigate the interplay between in-context learning attention heads and activation additions to develop context-invariant steering mechanisms.

6 Conclusion

In this paper, we introduced a mechanistic approach to aligning multi-agent behaviors in Large Language Models. By mapping the latent space geometry of the Prisoner's Dilemma, we pinpointed the locus of strategic intent at Layer 57. Our simple steering vector method successfully instills a generalized, zero-shot cooperative intent that preserves linguistic capabilities perfectly, outperforming baseline prompting and Representation Engineering. By explicitly outlining the limitations of this method against adversarial contexts, we chart a rigorous path forward for reliable activation engineering in AI alignment.

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